

Hospital Management System

DS-SE-2B - Team 5 - GCP Project Plan

Deadline: First week of May 2026

Team Overview

Member	Roll #	Role	Primary Modules
Zainab Farooq	25L-3048	Team Lead	Person base class, RBAC/Login, Staff module
Aliha Saeed	25L-3015	Patient module	Patient class, Medical records, PatientForm GUI
Zuha Shahzad	25L-3016	Doctor & appointments	Doctor class, Appointment logic, GUI forms
M. Mueez Ashfaq	25L-3054	Billing & records	Billing class, MedicalRecord form, Ambulance module
Shauzab Mukhtar	25L-3084	Inventory & rooms	Pharmacy class, Room management, GUI forms

Week 1 — Apr 18 to 24 | Planning & backend classes

All members together (first day — 2 to 3 hrs)

- Draw UML class diagram and agree on class interfaces
- Set up shared GitHub repo with branch-per-member strategy
- Create Visual Studio WinForms solution structure
- Agree on file naming, coding style, and fstream file paths

ZF	Zainab Farooq Team Lead	• Person base class (name, ID, contact — private attrs + getters/setters) • Login & RBAC system — role-based access for Admin/Doctor/Staff/Patient • users.txt file handling for authentication
AS	Aliha Saeed Patient module	• Patient class (inherits Person) — medical ID, diagnosis history • Patient registration, search, update logic • fstream save/load for patients.txt
ZS	Zuha Shahzad Doctor & appointments	• Doctor class (inherits Person) — specialization, availability • Doctor management logic (add, view, update) • Appointment class — scheduling, modification, fstream save/load
MA	M. Mueez Ashfaq Billing & ambulance	• Medical records class — diagnosis, prescriptions, treatment history • Billing class — invoice generation logic and fstream save/load • Ambulance/emergency class — dispatch, severity, transport logic
SM	Shauzab Mukhtar Inventory & rooms	• Pharmacy/inventory class — medicine stock, prescription dispensing • Room & bed management class — allocation, release, availability • fstream save/load for both modules

Checkpoint — Apr 24

- All backend classes compile and run correctly in terminal (no GUI yet)
- Each member demos their class: create an object, save to file, reload it
- Person inheritance chain confirmed working (Patient, Doctor, Staff)

Week 2 — Apr 25 to 30 | GUI forms & integration

ZF **Zainab Farooq**
Team Lead

- Staff class (inherits Person) — roles, shifts, assignments
- Main window layout — dashboard with role-based menu switching
- StaffForm GUI — add, view, manage staff
- AmbulanceForm GUI — dispatch and emergency coordination screen

AS **Aliha Saeed**
Patient module

- PatientForm — register, search, view patient list (DataGridView)
- MedicalRecordForm — view diagnosis history, add new record

ZS **Zuha Shahzad**
Doctor & appointments

- DoctorForm — add doctor, view schedule, manage availability
- AppointmentForm — book, modify, cancel appointments

MA **M. Mueez Ashfaq**
Billing & ambulance

- BillingForm — generate invoice, view payment status
- Ambulance backend: dispatch logic, severity levels, patient transport

SM **Shauzab Mukhtar**
Inventory & rooms

- PharmacyForm — stock view, add medicine, process prescriptions
- RoomForm — allocate bed, release, view availability grid

Checkpoint — Apr 30

- All Forms working individually and navigable from main dashboard
- Login screen functional — correct dashboard loads per role
- Merge all branches — Zainab reviews and approves pull requests
- Resolve conflicts in shared files (Person.h, main form navigation)
- End-to-end test: register patient → book appointment → generate bill

May 1 to 4 | Polish, bug fixes & demo prep

All members — final sprint tasks

- Fix any crash bugs found during full system testing
- Verify RBAC: each role sees only permitted screens and buttons
- Confirm data persistence: all records survive app close and reopen
- Clean up UI: fix spacing, labels, validation messages, error handling
- Each member prepares a 2-minute module walkthrough for demo
- Zainab: prepare OOP concepts summary (which class shows which concept)

Timeline at a glance

Phase	Progress	Target
● Backend done		Apr 24
● GUI done		Apr 28
● Merge & test		Apr 30
● Submission		May 1-4

Ground rules for the team

GitHub discipline	One branch per member. Never commit directly to main. Zainab approves all merges. This prevents one broken file from crashing everyone's work.
No logic in Forms	Keep button click handlers thin — they call your C++ class methods only. Business logic belongs in the class, not in the GUI. This keeps OOP clean.
Test before merging	Your module must compile and console-test before you open a pull request. Don't merge broken code and leave it for others to discover.
Merge day — Apr 30	Plan a 2 to 3 hour team call for the merge. Shared files like Person.h and the main form's navigation will have conflicts that must be resolved together, not solo.
Demo flow	Be ready to show: login with 3 roles, register a patient, book an appointment, generate a bill. That single flow touches every OOP concept in your proposal.